

$$\begin{aligned} \text{HD} \rightarrow & \frac{1}{16\pi^2} \left( \frac{2}{27} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_{\mathbf{d}}^{\mathbf{P}}] - \frac{5}{36} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{d}}^{\mathbf{P}}] + \right. \\ & \frac{8}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{d}}^{\mathbf{P}}] + \frac{2}{9} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_{\mathbf{e}}^{\mathbf{P}}] - \frac{5}{12} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{e}}^{\mathbf{P}}] + \\ & \frac{8}{45} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{e}}^{\mathbf{P}}] + \left( -c_{2\gamma} g_2^2 c_{\gamma}^2 \overline{y}_{\mathbf{e}}^{\text{pr}} y_{\mathbf{e}}^{\text{pr}} + \frac{1}{36} \sum_{\mathbf{p}} \left( 4 \mathbf{g}_1^4 + 9 \mathbf{g}_2^4 c_{2\gamma}^2 \right) \right) \text{LF}_{3,0}[\mathbf{m}_{\mathbf{l}}^{\mathbf{P}}] + \\ & \left( c_{2\gamma} g_2^2 c_{\gamma}^2 \overline{y}_{\mathbf{e}}^{\text{pr}} y_{\mathbf{e}}^{\text{pr}} - \frac{1}{24} \sum_{\mathbf{p}} \left( 5 \mathbf{g}_1^4 + 6 \mathbf{g}_2^4 c_{2\gamma}^2 \right) \right) \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{l}}^{\mathbf{P}}] + \frac{4}{45} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{l}}^{\mathbf{P}}] + \\ & \left( \frac{1}{27} \sum_{\mathbf{p}} \mathbf{g}_1^4 + \frac{3}{4} c_{2\gamma} g_2^2 \left( -4 c_{\gamma}^2 \overline{y}_{\mathbf{d}}^{\text{pr}} y_{\mathbf{d}}^{\text{pr}} + 4 s_{\gamma}^2 \overline{y}_{\mathbf{u}}^{\text{pr}} y_{\mathbf{u}}^{\text{pr}} + \sum_{\mathbf{p}} c_{2\gamma} g_2^2 \right) \right) \text{LF}_{3,0}[\mathbf{m}_{\mathbf{q}}^{\mathbf{P}}] + \\ & \left( 3 c_{2\gamma} g_2^2 \left( c_{\gamma}^2 \overline{y}_{\mathbf{d}}^{\text{pr}} y_{\mathbf{d}}^{\text{pr}} - s_{\gamma}^2 \overline{y}_{\mathbf{u}}^{\text{pr}} y_{\mathbf{u}}^{\text{pr}} \right) - \frac{1}{72} \sum_{\mathbf{p}} \left( 5 \mathbf{g}_1^4 + 54 \mathbf{g}_2^4 c_{2\gamma}^2 \right) \right) \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{q}}^{\mathbf{P}}] + \\ & \frac{4}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{q}}^{\mathbf{P}}] + \frac{8}{27} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_{\mathbf{u}}^{\mathbf{P}}] - \frac{5}{9} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_{\mathbf{u}}^{\mathbf{P}}] + \frac{32}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbf{u}}^{\mathbf{P}}] + \\ & \frac{1}{576} \left( \mathbf{g}_1^4 \left( 73 + 9 c_{4\gamma} \left( -2 + c_{4\gamma} \right) - 36 s_{2\gamma}^4 \right) + 18 \mathbf{g}_1^2 g_2^2 \left( -3 + c_{4\gamma} \left( 2 + c_{4\gamma} \right) - 4 s_{2\gamma}^4 \right) + \right. \\ & \quad 9 g_2^4 \left( \left( 3 + c_{4\gamma} \right)^2 - 4 s_{2\gamma}^4 \right) \left. \right) \text{LF}_{3,0}[\mathbf{m}_{\mathbb{H}}] + \frac{1}{192} \left( -6 \mathbf{g}_1^2 g_2^2 \left( -3 + c_{4\gamma} \left( 2 + c_{4\gamma} \right) - 4 s_{2\gamma}^4 \right) + \right. \\ & \quad 3 g_2^4 \left( -\left( 3 + c_{4\gamma} \right)^2 + 4 s_{2\gamma}^4 \right) + \mathbf{g}_1^4 \left( -43 - 3 c_{4\gamma} \left( -2 + c_{4\gamma} \right) + 12 s_{2\gamma}^4 \right) \left. \right) \text{LF}_{4,-1}[\mathbf{m}_{\mathbb{H}}] + \\ & \frac{4}{45} \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_{\mathbb{H}}] + \frac{1}{9} \mathbf{g}_1^4 \text{LF}_{3,0}[\tilde{\mu}] + \frac{1}{6} \mathbf{g}_1^4 \text{LF}_{4,-1}[\tilde{\mu}] - \frac{8}{45} \mathbf{g}_1^4 \text{LF}_{5,-2}[\tilde{\mu}] + \\ & \frac{1}{8} \mathbf{g}_1^4 \text{LF}_{2,2,-1}[\mathbf{m}_1, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 + \frac{1}{4} \mathbf{g}_1^4 m_1^2 \text{LF}_{2,2,0}[\mathbf{m}_1, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 + \\ & \frac{17}{8} g_2^4 \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 + \frac{1}{4} g_2^4 m_2^2 \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 - \\ & 4 g_2^4 \text{LF}_{3,2,-2}[\mathbf{m}_2, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 + 3 g_2^4 m_2^2 \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 + \\ & 2 g_2^4 \text{LF}_{3,3,-3}[\mathbf{m}_2, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 - 2 g_2^4 m_2^2 \text{LF}_{3,3,-2}[\mathbf{m}_2, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 + \\ & 2 g_2^4 \text{LF}_{4,2,-3}[\mathbf{m}_2, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 - 2 g_2^4 m_2^2 \text{LF}_{4,2,-2}[\mathbf{m}_2, \tilde{\mu}] \left( c_{\gamma}^2 - s_{\gamma}^2 \right)^2 + \\ & \frac{2}{3} g_1^2 \left( c_{\gamma} \overline{a}_{\mathbf{d}}^{\text{pr}} - s_{\gamma} \tilde{\mu} \overline{y}_{\mathbf{d}}^{\text{pr}} \right) \left( c_{\gamma} a_{\mathbf{d}}^{\text{pr}} - s_{\gamma} \tilde{\mu} y_{\mathbf{d}}^{\text{pr}} \right) \text{LF}_{2,2,0}[\mathbf{m}_{\mathbf{d}}^{\mathbf{r}}, \mathbf{m}_{\mathbf{q}}^{\mathbf{p}}] - \\ & \frac{1}{3} g_1^2 \left( c_{\gamma} \overline{a}_{\mathbf{d}}^{\text{pr}} - s_{\gamma} \tilde{\mu} \overline{y}_{\mathbf{d}}^{\text{pr}} \right) \left( c_{\gamma} a_{\mathbf{d}}^{\text{pr}} - s_{\gamma} \tilde{\mu} y_{\mathbf{d}}^{\text{pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_{\mathbf{d}}^{\mathbf{r}}, \mathbf{m}_{\mathbf{q}}^{\mathbf{p}}] + \\ & \frac{2}{3} g_1^2 \left( c_{\gamma} \overline{a}_{\mathbf{e}}^{\text{pr}} - s_{\gamma} \tilde{\mu} \overline{y}_{\mathbf{e}}^{\text{pr}} \right) \left( c_{\gamma} a_{\mathbf{e}}^{\text{pr}} - s_{\gamma} \tilde{\mu} y_{\mathbf{e}}^{\text{pr}} \right) \text{LF}_{2,2,0}[\mathbf{m}_{\mathbf{e}}^{\mathbf{r}}, \mathbf{m}_{\mathbf{l}}^{\mathbf{p}}] - \\ & \frac{1}{3} g_1^2 \left( c_{\gamma} \overline{a}_{\mathbf{e}}^{\text{pr}} - s_{\gamma} \tilde{\mu} \overline{y}_{\mathbf{e}}^{\text{pr}} \right) \left( c_{\gamma} a_{\mathbf{e}}^{\text{pr}} - s_{\gamma} \tilde{\mu} y_{\mathbf{e}}^{\text{pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_{\mathbf{e}}^{\mathbf{r}}, \mathbf{m}_{\mathbf{l}}^{\mathbf{p}}] - \\ & \frac{2}{3} g_1^2 \left( c_{\gamma} \overline{a}_{\mathbf{e}}^{\text{pr}} - s_{\gamma} \tilde{\mu} \overline{y}_{\mathbf{e}}^{\text{pr}} \right) \left( c_{\gamma} a_{\mathbf{e}}^{\text{pr}} - s_{\gamma} \tilde{\mu} y_{\mathbf{e}}^{\text{pr}} \right) \text{LF}_{3,1,0}[\mathbf{m}_{\mathbf{l}}^{\mathbf{p}}, \mathbf{m}_{\mathbf{e}}^{\mathbf{r}}] - \\ & \frac{2}{3} g_1^2 \left( c_{\gamma} \overline{a}_{\mathbf{e}}^{\text{pr}} - s_{\gamma} \tilde{\mu} \overline{y}_{\mathbf{e}}^{\text{pr}} \right) \left( c_{\gamma} a_{\mathbf{e}}^{\text{pr}} - s_{\gamma} \tilde{\mu} y_{\mathbf{e}}^{\text{pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_{\mathbf{l}}^{\mathbf{p}}, \mathbf{m}_{\mathbf{e}}^{\mathbf{r}}] + \\ & c_{\gamma}^4 \overline{y}_{\mathbf{e}}^{\text{pr}} \overline{y}_{\mathbf{e}}^{\text{st}} y_{\mathbf{e}}^{\text{pt}} y_{\mathbf{e}}^{\text{sr}} \text{LF}_{2,1,0}[\mathbf{m}_{\mathbf{l}}^{\mathbf{p}}, \mathbf{m}_{\mathbf{l}}^{\mathbf{s}}] - c_{\gamma}^4 \overline{y}_{\mathbf{e}}^{\text{pr}} \overline{y}_{\mathbf{e}}^{\text{st}} y_{\mathbf{e}}^{\text{pt}} y_{\mathbf{e}}^{\text{sr}} \text{LF}_{3,1,-1}[\mathbf{m}_{\mathbf{l}}^{\mathbf{p}}, \mathbf{m}_{\mathbf{l}}^{\mathbf{s}}] - \\ & \frac{2}{3} g_1^2 \left( c_{\gamma} \overline{a}_{\mathbf{d}}^{\text{pr}} - s_{\gamma} \tilde{\mu} \overline{y}_{\mathbf{d}}^{\text{pr}} \right) \left( c_{\gamma} a_{\mathbf{d}}^{\text{pr}} - s_{\gamma} \tilde{\mu} y_{\mathbf{d}}^{\text{pr}} \right) \text{LF}_{3,1,0}[\mathbf{m}_{\mathbf{q}}^{\mathbf{p}}, \mathbf{m}_{\mathbf{d}}^{\mathbf{r}}] - \\ & \frac{2}{3} g_1^2 \left( c_{\gamma} \overline{a}_{\mathbf{d}}^{\text{pr}} - s_{\gamma} \tilde{\mu} \overline{y}_{\mathbf{d}}^{\text{pr}} \right) \left( c_{\gamma} a_{\mathbf{d}}^{\text{pr}} - s_{\$$